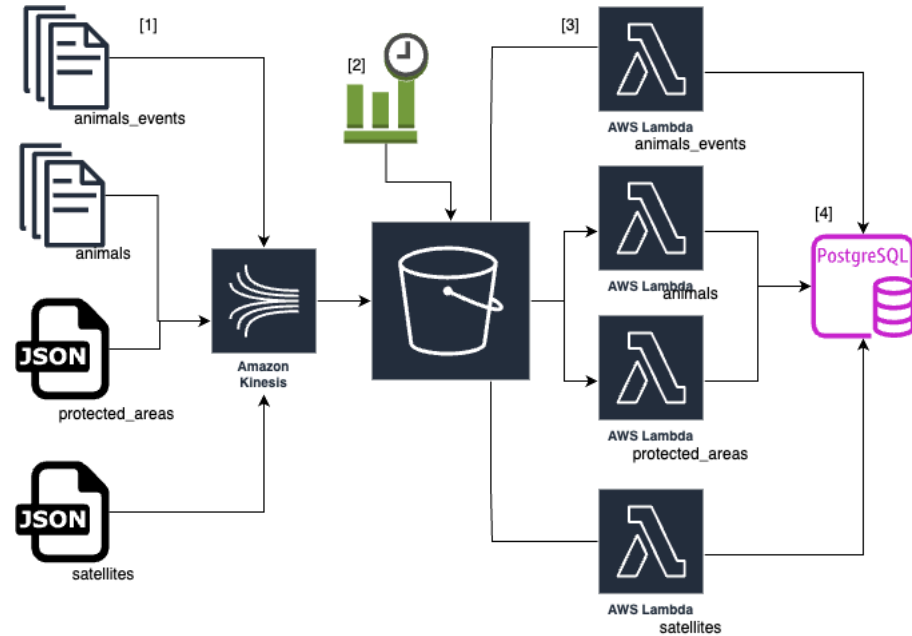


ML assessment design doc

Flow diagram



Description

TLDR

This ETL:

- **Ingestion:** Uses Amazon Kinesis for live-streamed animal tracking data.
- **Transformation:** AWS Lambda converts data into GeoJSON format.
- **Storage:** Data stored in Amazon RDS PostgreSQL with PostGIS; raw data in Amazon S3.
- **Automation:** Event-driven triggers and scheduled tasks maintain continuous data flow.

[1] DATA INGESTIONS

- The ingestion process begins with Amazon Kinesis Data Streams capturing all the real-time data from monitoring services.
- The data then is stored as raw in s3. This files are JSON as it is how kinesis send data
- I would keep it there for one month and then send it to cold storage [S3 glazier]
- This process leverages Kinesis's scalability and Lambda's serverless capabilities for efficient real-time data handling.

[2] AUTOMATIZATION

- AWS Lambda functions are triggered by EventBridge or Lambda S3 event triggers
- EventBridge will handle `animals_events` which the dataset that we expect to change and increase over time
- S3 event triggers would work better for the rest of datasets as we do not expect them to change much over time
- Retry mechanisms and Dead-Letter Queues (DLQs) capture failed events for analysis and reprocessing.

[3] TRANSFORMATIONS

- Lambda functions can handle up to 10gb of data and have a processing time of 15 minutes which is enough to perform necessary transformations
- Each data will have their own lambda as each one requires different transformation
 - animals
 - Missing values
 - Formats
 - animals events
 - Missing values
 - Data format
 - Check timestamp to be correct
 - Create a geometry column with the coordinates
 - Transform to geojson
 - satellites
 - Get geometry with bounding_box
 - Transform to geojson
 - protected_areas
 - Transform to geojson
- Lambda functions will store the data in a Postgres database

[4] STORAGE

- I will storage GeoJSON data in Amazon RDS PostgreSQL with PostGIS for geospatial capabilities.
- I will use indexing and partitioning in PostgreSQL for enhanced query performance.
- Schemas will include a `jsonb` column for efficient GeoJSON object storage and querying.
- We will use S3 for raw data

OTHERS

- **Monitoring:** Amazon CloudWatch tracks metrics (data ingestion rates, Lambda execution times, error rates) with configured alarms.
- **Redundancy:** Kinesis configured with multiple shards and data replication to prevent data loss.
- **Error Handling:** AWS Lambda includes retry logic and DLQs for managing transient failures.
- **Security:** Implements IAM role restrictions and data encryption (at rest and in transit).

POTENTIAL ENHANCEMENTS AND ALTERNATIVE TOOLS

- Ingestion: If the volume of data increases we can change lambda functions for EMR or AWS glue to handle big data
- Orchestration: We could use AWS step functions or Apache Airflow if the transformations get more complex